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1. From Observation to Theory

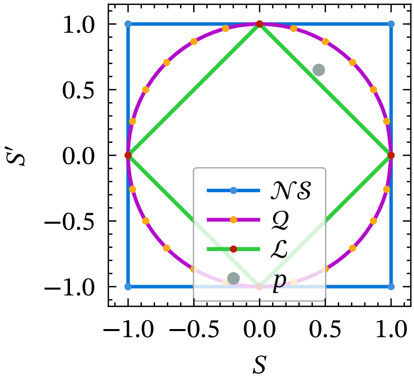
1.1. *The Laboratory*

A theory is an **explanation** of an experiment.

$$\xrightarrow{\text{results}} (v_1, v_1, v_2, v_3) \xrightarrow{\text{statistics}} \quad p = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{4} \\ \frac{1}{4} \end{pmatrix}$$

In 1982, Alain Aspect's experiments observed correlations that cannot be explained by any local (classical) theory. [Bel64]

Experimental correlations live between two natural bounds: the classical set $\mathcal{L}$ and the non-signaling set $\mathcal{NS}$.



Let t such that $|N|_1 = 4t$ for technical reason.

Quantum theory, formulated with Hilbert spaces, are formulated with $t \to \infty$.

- For example, with only **4 experimental runs**, it may be impossible to realize a statistic $p \in \mathcal{Q}$.
- At this finite scale: either $p \in \mathcal{L}$ or may even fall outside $\mathcal{NS}$.
- Our goal is to develop a **constructive approach**: for each finite value of t, we want to describe which empirical models are possible and which are not.

$$\xrightarrow{\text{results}} (v_1, v_1, v_2, v_3) \xrightarrow{\text{countings}} N_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \to N_2 = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} \to N_3 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \to N_4 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$$

We want to observe the trajectory by which a correlation p is progressively built.

Why does this **create a lattice**? Since $p = \frac{N}{t}$, for example with $t = 4$, there are only 4 possibles correlations for p.

We need to give a **meaning** to all the intermediate experiments, whereas the usual quantum formalism gives meaning mainly to the limiting statistics.

2. Dynamic Automata

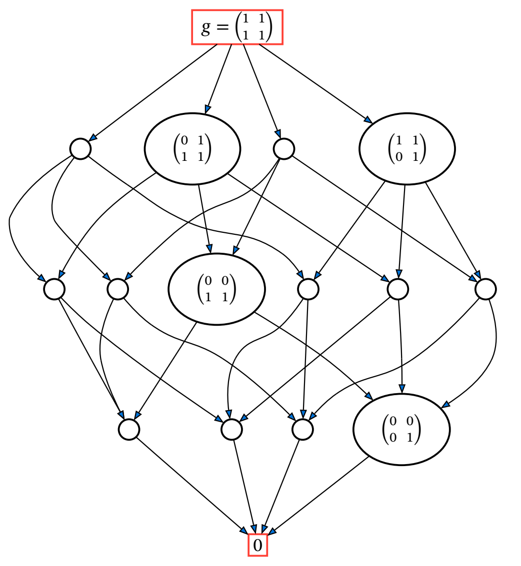
What happens during the production of this final counts N.

A **generator family** $\mathcal{G} \subseteq \mathbb{N}^V$ is the elementary explanations of our theory.

Example of $\mathcal{G}$:

$$g_{12} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}; g_{33} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}; g_{42} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix}; \dots$$

How to build g ? What is the meaning of **being doing** g.



A **residual state** of g is an element R such that $0 \leq R < g$.

Open
residual $R > 0$

Closed
residual $R = 0$

We can express observation with a sum of elementary explanation $N = \begin{pmatrix} 1 & 2 & 2 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix} = g_{41} + g_{12} + g_{13}$

We can express observation with **residual state** :

$$N = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix} = g_{41} = \left(g_{11} - \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \right) + \left(g_{13} - \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \right)$$

We view an observation as **something we are in the process of producing**.

2.0.1. Dynamic States

$$g_1 \to \dots \to (g_1, R_1) \to \dots \to 0$$

$$g_1 \to \dots \to (g_1, R_2) \to \dots \to 0$$

$$g_2 \to \dots \to (g_2, R_3) \to \dots \to 0$$

Let $\eta = \mathbb{O}_\eta := \{\{(g, R) \mid g \in \mathcal{G} \wedge 0 \leq R \leq g\}\} = (c_\eta, O_\eta)$ an multiset of intermediate automaton state, η **explain** N if :

$$N = \sum_{(g,R) \text{ in } \mathbb{O}_\eta} g - R = \sum_{g \in \mathcal{G}} c_\eta(g) \cdot g + \sum_{(g,R) \text{ in } O_\eta} g - R$$

The explanation set of visible count N is $\mathcal{H}_\mathcal{G}(N) := \{\eta \mid N_\eta = N\} = \left\{\mathbb{O} \mid \sum_{(g,R) \text{ in } \mathbb{O}_\eta} g - R = N\right\}$

The **completed future** of η is $F_\eta = \sum_{(g,R) \text{ in } \mathbb{O}_\eta} g = N_\eta + \sum_{(g,R) \text{ in } O_\eta} R$

The **stabilized past** of η is $P_\eta = \sum_{(g,0) \text{ in } \mathbb{O}_\eta} g = \sum_{g \in \mathcal{G}} c_\eta(g) \cdot g$

2.0.2. The Signature of an Interruption

Past
P_η: closed explanation

Visible
N_η: observed count

Future
F_η: completed count

2.1. *What append in the middle?*

Imagine, the lab perform the experimentation to get N_2, but was interrupt in the middle. Let N_1 the intermediate result.
Example :

$$\begin{pmatrix} 4 & 3 \\ 8 & 0 \end{pmatrix} \leq \begin{pmatrix} 9 & 3 \\ 8 & 6 \end{pmatrix}$$

We need two explanation, η_1 for N_1 and η_2 for N_2 such as the two was indeed the intermediate automaton state of the **same process**.

Transition condition $\eta_1 \rightsquigarrow \eta_2$ is define such that any generator that will open, will either **continue**, or **close**.

2.2. *The Theory* $\mathbb{T} = (\mathcal{G}, \xi)$

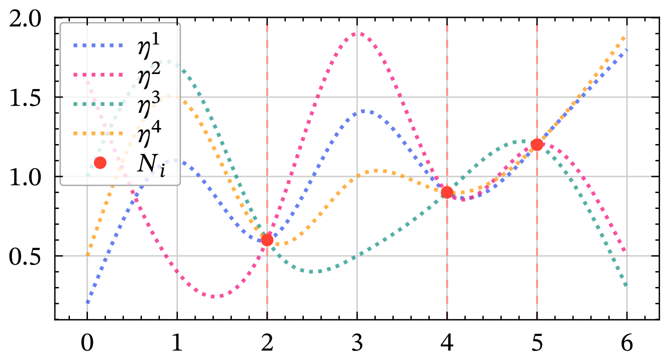
The model has two ingredients.

Structure: $\mathcal{G}$
Which elementary generators are allowed?

Dynamics: ξ
Which trajectories are admissible?

The theory explains a sequence of observations when there exists at least one compatible hidden trajectory:

$N_{1:k} \in \mathbb{T} \Leftrightarrow \mathcal{H}_{\mathcal{G},\xi}(N_{1:k}) := \{\eta_{1:k} \mid \forall i \in [k]\ N_{\eta_i} = N \wedge \eta_i \rightsquigarrow \eta_{i+1} \wedge \xi \vDash \boldsymbol{\eta_{1:k}}\} \neq \varnothing$.

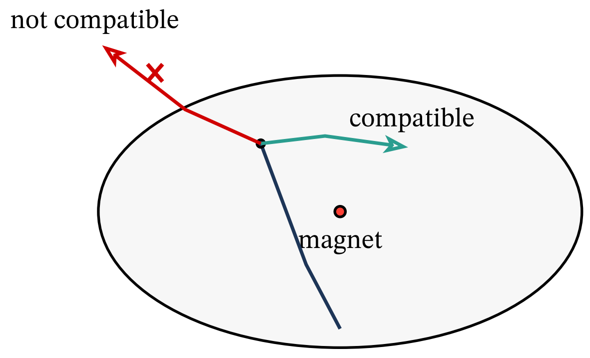


In this example : $\mathcal{H}_{\mathcal{G},\xi}(N_{1:3}) = \{\eta^1_{1:3}, \eta^2_{1:3}, \eta^3_{1:3}, \eta^4_{1:3}\}$

Example of stability constraint ξ that controls open generators:

$$o_\eta(g) \leq \lambda c_\eta(g) + B$$

The constraint behaves like a magnet.
It does **not choose the path**, but it force the dynamics of the process, it **filter**.



This is a constructive-theory viewpoint:

Not predetermined trajectories, but constrained possible trajectories.

3. What this Framework Adds

3.1. *Interruption Profiles*

Let's take $\mathcal{G}$ like before. Example $g_{11} = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$.

If an observation connot be express as the direct sum of close generator, it call **non-contextual**.

$$\text{Example: } \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The next idea is to describe the **internal dynamics** of a **trajectory** η through its interruption profile.

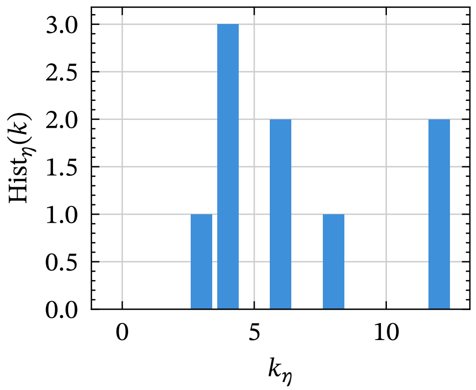
For a fixed explanation η, compare each coordinate separately:

$$P_\eta = \begin{pmatrix} 2 & 2 & 1 & 1 \\ 3 & 3 & {\color{blue}1} & 1 \\ 2 & 2 & {\color{red}3} & 3 \\ 1 & 1 & 3 & 3 \end{pmatrix} \to N = \begin{pmatrix} 2 & 2 & 2 & 3 \\ 4 & 5 & {\color{blue}2} & 2 \\ 5 & 4 & {\color{red}4} & 3 \\ 1 & 1 & 3 & 3 \end{pmatrix} \to F_\eta = \begin{pmatrix} 2 & 2 & 4 & 4 \\ 6 & 7 & {\color{blue}5} & 4 \\ 5 & 6 & {\color{red}4} & 3 \\ 1 & 1 & 3 & 3 \end{pmatrix}$$

$$\xrightarrow{\quad {\color{blue}2} \quad} {\color{red}4}$$

$$\text{time}_\eta = \frac{k_\eta}{12} = \frac{\begin{pmatrix} - & - & 4 & 8 \\ 4 & 6 & 3 & 4 \\ 12 & 6 & 12 & - \\ - & - & - & - \end{pmatrix}}{12}$$

We can measure how synchronized the events are along the evolution of an explanatory trajectory η.



$$\text{Hist}_\eta(k) = \#\{v \in V \mid k_\eta(v) = k\}$$

Then we can **filter** explanations using ξ, depending on the kind of explanation we want: **highly synchronized**, **uniform**, or something else.
Example: Causal constraint

$$k_\eta(v) \leq k_\eta(v') \quad vv' \in A$$

This is where the framework becomes genuinely dynamic: **constraints** can act on the **organization of the process**, not only on the final table.

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4. CONCLUSION

What will you learn?

- The framework on contextuality through **automata**.
- Why keeping integer counts allows us to express more structure, such as **trajectories** instead of only points.

What will we not cover in this presentation?

- How the internship evolved over time.
- How my dynamic automaton framework connects with the foundational papers on contextuality [ABM16, AB11], or with the hypergraph approach of [CSW14].
- How this framework could contribute to the broader study of quantum contextuality.
- Why this approach seems promising for understanding quantum phenomena.

Open questions:

- Can we find a theory $\mathbb{T} = (\mathcal{G}, \xi)$ that separates quantum correlations from classical and non-signaling correlations?
- Can we evaluate such a theory on observations such as $P_v = vP_{\mathrm{PR}} + (1 - v)P_{\mathrm{uni}}$, or more generally using [SWC25]?
- Can we test the framework on multipartite games such as GHZ, which are strongly contextual?
- Can we build a dynamic framework that progressively constructs itself?

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5. Annexe 1

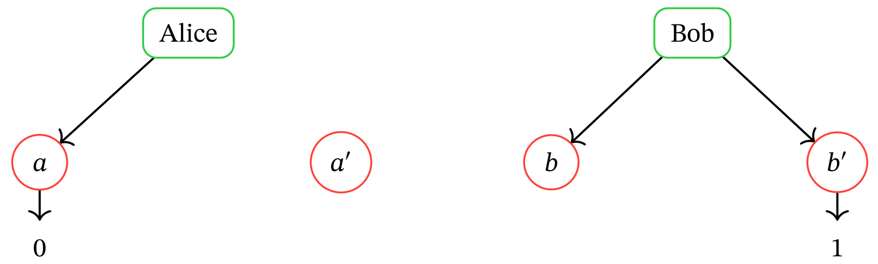
5.1. The Laboratory

We want to create a theory of theory, and a theory start by an **experience**.

In the Bell $(2, 2, 2)$ scenario:

A set of *measure* $X = \{a, a', b, b'\}$, a set of *outcome* $O = \{0, 1\}$.

Alice can choose a or a'. Bob can choose b or b'.
Example of context : $C = \{a, b\}C = \{a, b'\}$



An experiment first choose one context and get the result $s = (a \mapsto 0, b' \mapsto 1)$.

The *measurement context* is $\mathcal{M} = \{\{a, b\}, \{a, b'\}, \{a', b\}, \{a', b'\}\} \subset \mathcal{P}(X)$.

A local *section* $s \in \mathcal{E}(C)$, example $s = (a \mapsto 0, b' \mapsto 1)$.

We define the *visible event* set by $V \coloneqq \{(C, s) \mid C \in \mathcal{M}, s \in \mathcal{E}(C)\}$

It is what is actually recorded.

5.1.1. From Local Events to Contextuality

We also ask whether the result would have been the same in another context?

If Bob had chosen b instead of b', would Alice still have obtained $a \mapsto 0$?

Classically, the answer should be no: every measurement already has a value.

A *global section* is $g \in \mathcal{E}(X)$, i.e. a map between every measure and outcome :

$$g = (a \mapsto 0, a' \mapsto 1, b \mapsto 0, b' \mapsto 0)$$

For each context, the local result is only a restriction:

$$C = \{a, b\} \quad s = g|_C = (a \mapsto 0, b \mapsto 0)$$

A global section means: all counterfactual outcomes are already written.

5.2. Counts Before Probabilities

One run is not an experiment. An experiment is a collection of runs.

$C = \{a, b'\}$ $\xrightarrow{\text{run the experiment}}$ $s = (a \mapsto 0, b' \mapsto 1)$ $\xrightarrow{\text{record}}$ $N(C, s) = N(v) \in \mathbb{N}^V$

Instead of normalizing immediately, we keep the integer counts:

An *integer model* is $N \in \mathbb{N}^V$

After normalization $N_1 = \binom{v_1}{v_2} = \binom{1}{0}$ and $N_{1000} = \binom{v_1}{v_2} = \binom{1000}{0}$ both become probability $\binom{1}{0}$.

Before normalization They carry different levels of evidence. We are more confidence in N_{1000} instead of N_1.

$N \in \mathbb{N}^V$ is compatible if $\forall C, D \in \mathcal{M}$ two contexts and $u \in \mathcal{E}(C \cap D)$:

$$\sum_{s \in \mathcal{E}(C), s_{|C \cap D} = u} N(C, s) = \sum_{s \in \mathcal{E}(D), s_{|C \cap D} = u} N(D, s)$$

With $C = \{a, \mathbf{b}\}$ and $D = \{\mathbf{b}, c\}$ we need to have :

For $u = (\mathbf{b} \mapsto 0) : N(C, (a \mapsto 0, \mathbf{b} \mapsto 0)) + N(C, (a \mapsto 1, \mathbf{b} \mapsto 0)) = N(D, (\mathbf{b} \mapsto 0, c \mapsto 0)) + N(D, (\mathbf{b} \mapsto 0, c \mapsto 1))$

For $u = (\mathbf{b} \mapsto 1) : N(C, (a \mapsto 0, \mathbf{b} \mapsto 1)) + N(C, (a \mapsto 1, \mathbf{b} \mapsto 1)) = N(D, (\mathbf{b} \mapsto 1, c \mapsto 0)) + N(D, (\mathbf{b} \mapsto 1, c \mapsto 1))$

The marginal coincide on the common contexts. This call **non-signaling**, no information transit between the context (between Alice and Bob). We write this condition $\delta N = 0$

Each context can have him own number of event $|N_C|$, by applying the **non-signaling** condition, this force each context to have the same number of element $\forall C \in \mathcal{M} \quad |N_C| = t$.

$$A_{\mathcal{M}} N = t\mathbf{1} \qquad \leftarrow \text{same level}$$

$$\delta N = 0 \quad \leftarrow \text{non-signaling constrain}$$

This describe a cone $\mathcal{S}_{\text{ns}} = \{(N, t) \mid A_{\mathcal{M}} N = t\,\mathbb{1} \wedge \delta N = 0\} = tG_{\text{ns}} \cap \mathbb{N}$

5.2.1. Noncontextual Countings

A deterministic global section g gives one counting vector d_g.

An integer counting is *noncontextual* if it decomposes as a sum of such global sections:

$$\exists c : \mathcal{G} \to \mathbb{N}, \quad N = \sum_{g \in \mathcal{E}(X)} c_g d_g$$

Contextuality begins when N is compatible but no such decomposition exists.

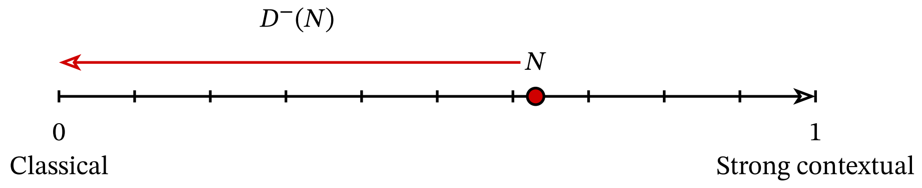
$N \in \mathcal{S}_{\text{ns}}$ but $N \notin \mathcal{S}_{\text{nc}}$.

5.3. The Quantum Tension

We have two thing, first quantum lives strictly between non-contextual (global section) and the all set $\mathcal{S}_{\text{ns}}$ of non-signaling model.

Second we can try to classify observation with what we call **contextual fraction** (in our case, because we have integer instead of probabilities we have minimal distance).

Let define $D^-(\hat{N}) = \min_{N \in \mathcal{S}_{\text{ns}}} |N - \hat{N}|_1$ the minimal distance with the closest explanation of the model $\mathcal{S}_{\text{ns}}$.



The farther N is from the classical side, the more contextual it is.

But, strong contextual not implies quantum or not, we cannot express quantumness by a distance with the classicality (GHZ)

We believe that quantum cannot be found if we think in term of point, or state, we need to analyse the trajectory!

How to formalise that?

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6. ANNEXE 2

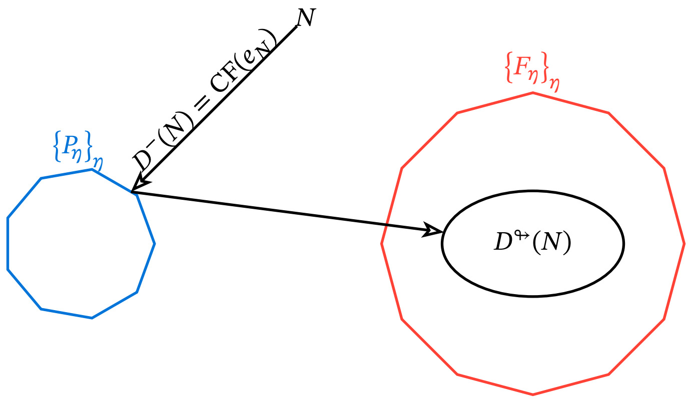
6.1. *Static Measure and Dynamic Enrichment*

How much of N can be explained by global sections?

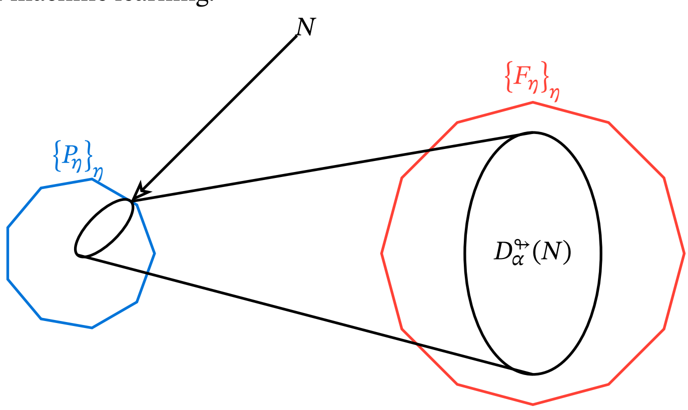
$D^{-}(N) \coloneqq \min_{\eta} |N - P_{\eta}|_1$ is equivalent to contextual fraction.

But the dynamic model also provide another direction, the completion distance.

$D^{\rightsquigarrow}(N) \coloneqq \min\{|F_{\eta} - N|_1 \mid d^{-}_{\eta} = D^{-}(N)\}$.



But we want a **dynamic** theory: one that can revisit its own axioms during the experiment and update them.
To make this possible, we relax the stabilized past, allowing the theory to explore alternative completions, in a way reminiscent of machine learning.



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